

2024 Swiss Camp and JAR AWS maintenance and GNSS efforts

18-24 August, 2025

These notes supplement the maintenance sheets

JEB, RABNI, pilot Kristian Höglund

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Who

1. Jason Box (JEB), Rasmus Nielsen (RABNI) from GEUS
2. Pilot was ____

What

While me and RABNI could only fly day 1 or 2, we got all done but surveying the newly installed stake triangle around SWC_O

Our day 1 was from 9 AM to 7 PM, windy but otherwise warm 2°C

Here are [JEB photos](#).

We expected to fly day 2... but clouds blanketed Ilulissat all day. It's damn good business we had a very pragmatic and wise pilot who (to be honest) pushed us to do all we could day one. He had a better sense that the 50/50 odds of 'weather' day 2 meant we may be unlucky. I keep re-learning to 'make hay when the sun is shining', i.e., push hard when you are out flying even if u get tired. And we were getting tired and hungry after 7 hours, but pushed another 3 hours of work... getting up to SWC_O and fixing all there... if I weren't tired I think I would have thought

to run the Javad GNSS to survey the SWC_O triangle, then we would be at 100% success of objectives. We ended up having to settle for installing the triangle and get SMB from it on revisit next year, just no precise positioning of the JAR triangle stakes. We DID get GNSS to survey the JAR_O triangle, so Manfred will also be happy.

For some reason, I'm most satisfied we got the SWC_O T string working. And at JAR_O, of course it was getting the PTA in, that we hand carried to CPH. Bringing the extra splitter was key because either I damaged it on site during trouble shoot or it was already un-soldered. Then I learned a new trick which was to download the onboard program PROMICE v3 rewiderrd.1.5 and push that back onto the logger, which made the PTA and SR-50s work, were giving NaNs before. Strange. But it worked. As I wrote to someone, I think it was RSF, my theory has to do with the .stn file that maps the public channels with the variable names in the .cr1 file. So, I concluded it was not a problem with the onboard CRBasic, but the damn .stn that's more about the device stored on the laptop mapping the public variables correctly. I won't forget that theory in the future. It may actually be the solution to the QAS_L issue me and Rasmus faced in April. It's not, in my theory, necessary to bring a new logger box! Anyway, we're now well prepared for how to get PTA working at QAS_L 2 or 5 Sep this year.

Notes

- GNSS info in [lab-folder](#) under projects and named SWC_JAR_GNSS_setup. Here is a link, [SWC_JAR_GNSS_setup](#)

Main accomplishments

1. Success in just 1 charter day of work. Charter day 2 was no fly.
 - a. Got PTA re-installed at JAR_O
 - b. T-string re-installed at SWC_O
 - c. SR50 replacement for boom and stake **except boom at SWC_O**
 - d. Replaced Lufft, failed spring 2025
 - e. Obtained 10 min and 60 min data
2. Ablation estimates

- a. JAR_O
 - i. From maintenance sheets...2024 08 19 438.5 cm, 2025 08 20 318 cm
Difference: $438.5 - 318 = 120.5$... makes sense
 - ii. From daily stake SR50 data: no result as SR50 was out
 - b. SWC_O
 - i. From maintenance sheets...2024 08 20 182 cm, 2025 08 20 437 cm
Difference: $437 - 182 = 255$... seems too high, subtracting 2 m makes more sense, i.e. 55 cm ablation
 - ii. From daily stake SR50 data: end of melt season (EOMS) 2024 to EOMS 2025 = 0.565 m
3. The existing *continuously operating reference station* (CORS) GNSS had no data on them, as half expected. But we set in the new boards and expect the site is recording

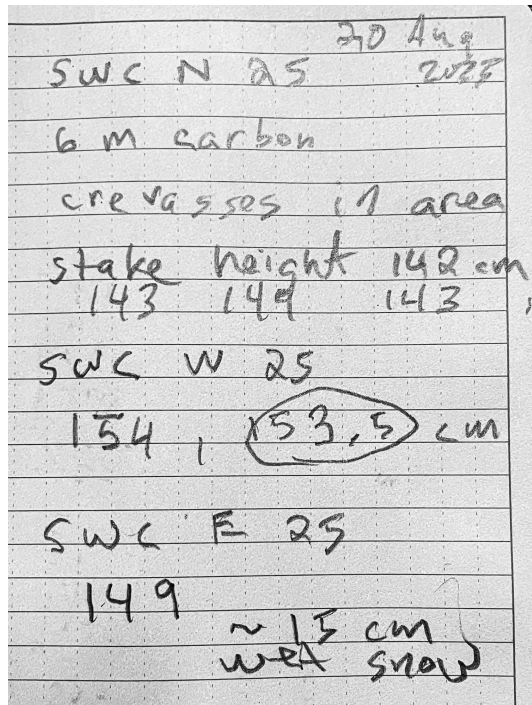
Triangles around the AWS, survey 20 August 2025



It's a geodetic survey...The green GNSS antennae on the top of the carbon fiber stake, drilled 6 m into the ice, is recording for 15 minutes. With the data, we obtain a precise (to 1 cm) position on the Greenland ice sheet. Each year, we make that measurement at the same stake and from the two positions, we get the ice movement in 3D, roughly 20 cm per day horizontally and 2 m downward in the vertical. We have 3 such stakes in a triangle around our weather stations. In the center, we are measuring the position every 3 hours and so we get the displacement of the weather station and the triangle of surrounding points. Yes, it's a rather complicated measurement. The point is to continue what M. Stober started in 1991, each 1-2 years surveying that kind of triangle. In each year of repeat survey, we measure the surface and total mass change for a triangular volume.

- JAR 25N
 - Stake top cm: 141, 133.5, 140, 137 what is the height of the stake when we arrived?
 - removed the upper 1 of 3 stakes
 - Then we put the green antenna on the carbon
 - The bottom of the antenna the brass part is 140 cm above ice
 - We started the Javed recording at 1409 UTC
 - Antenna cable, pointing 205°, south
 - The surrounding surface has a stream that is about 6 m west of the stake and about 300 m to the southeast is a lake and we can see the weather station it's up arise in ice approximately 20 m higher than this position. This northern state is got higher elevations everywhere except to the southeast lake
 - Stopping the recording at 1324 local time
 -
- JAR 25W
 - At the west site we removed 2 m from the stake and what was left was 181, 184, 185, need to 182
 - Add to centimeters to get the bottom of the antenna
 - At this West location there is only 2 m of the stake remaining in ice and we stop the recording at 1353 local time
 - removed the upper 1 of 3 stakes
- JAR 25E
 - carbon top at 210, 213, 211.5, 215
 - For the survey, add 2 cm for the bottom of the antenna
 - 15 min recording started at 1403 local
 - Did not remove the upper 1 of 3 stakes

- SWC_O
 - GNSS replacement
 - See maintenance sheet
- Installation and survey of 3 survey stakes surrounding SWC_O



To do next time

- Confirmation CORS was recording at both JAR_O and SWC_O
- Replace SR50 on SWC boom
- Removal of SWC and JAR AWS installed in 2020.
- Rent 'african' Mitsubishi instead of red HiAce

Things we learned

JAR_O

- Stake height on arrival 19 Aug 2024 was 438 cm, 3 Aug 2022 256.5, suggesting average 90.75 cm per year ice ablation for the 2 melt seasons (2023, 2024). CHECK the following:

Using the following, the EOMS 2022/23 would be 1.298 m and the 2023/24 would be 0.985 m

- Ice ablation EOMS 2023/24 from daily z_stake: 0.5215 m using the following code
- `df['temp']=df.z_stake.copy()`
- `df.loc['2024-08-20:', 'temp'] += 2.451`
- `df.loc['2023-09-19:', 'temp'] -= 0.9852`
- `# df.loc['2024-08-17:', 'z_stake'] -= 2`
- `print(df.loc['2023-08-10':'2023-11-10', 'temp'].max())`
- `print(df.loc['2024-08-1':'2024-11-10', 'temp'].max())`

Ilulissat accommodations and logistics

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Car rental, clunky large but effective red Toyota HiAce

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